A Multi-modal Hybrid Tracking System Using Magnetic Localization, Inertial Sensing, and Fiducial Vision for Low-Cost Robot Teleoperation

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Abstract

This paper presents the design, implementation, and validation of a multi-modal hybrid tracking system that fuses magnetic localization, inertial measurement, and fiducial vision to generate real-time end-effector commands for robot teleoperation in simulation. The system employs a deliberate ownership decomposition: a 3x3 magnetometer array estimates board-local magnet motion via constrained magnetic dipole inversion, an inertial measurement unit (IMU) provides startup-referenced orientation change, and an AprilTag vision subsystem supplies global translation. A hybrid combiner composes these three channels into a unified Franka Panda end-effector target within NVIDIA Isaac Sim at 120 Hz. The architecture intentionally removes the AprilTag orientation estimation before fusion so that the IMU remains the sole signal for orientation to avoid complications. Hardware is built around an STM32WB55-based BLE wristband with an LSM6DSO IMU and nine LIS3MDL magnetometers multiplexed via a TCA9548A I2C switch. Because the consolidated PCB exhibited an unresolved fault on the magnetometer-extension side, the demonstrated system runs on a deliberate bench-prototype configuration that reuses the working half of the previous PCB together with the functional extension board; this preserves the full sensing pipeline while bypassing the faulty interconnect, the tradeoff is the larger size and potential blockage of vision between the camera and the AprilTag cube due to exposed wires. Experimental validation confirms that the ownership-based fusion law correctly separates translational and rotational contributions, maintains stability through transient sensor dropouts, and enables smooth teleoperation of a simulated manipulator from a wearable device. Planned near-term work includes fixing the consolidated PCB and replacing the bulky AprilTag cube with an overhead camera observing a small circular WhyCon fiducial mounted directly on the wristband, substantially reducing the size of the wearable while preserving the translation-ownership role. Together the results demonstrate the feasibility of low-cost, accessible teleoperation with multi-modal sensing using commodity hardware.

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1. Introduction

Teleoperation enables human operators to control remote robotic systems in environments that are hazardous or inaccessible, with applications spanning surgical robotics, industrial

manipulation, and hazardous-material handling [1]. While significant research effort has been directed toward reducing communication latency in teleoperation [1][2], the integration of intuitive, multi-modal sensing for operator input remains a challenge — particularly in low-cost systems. State-of-the-art surgical platforms such as the da Vinci system did not incorporate haptic feedback until the da Vinci 5's introduction in 2024 [3], and tactile feedback in robot-assisted surgery remains largely confined to research settings [5]. Power constraints [4], integration difficulties [5], and cost [6] continue to limit the deployment of haptic-capable, accessible teleoperation setups.

A parallel challenge is the development of accurate, wearable pose-tracking devices that can serve as natural master devices for teleoperation. Existing approaches include optical motion capture, electromagnetic tracking, and inertial navigation, each with trade-offs in cost, accuracy, and environmental constraints. Magnetic tracking using permanent magnets and magnetometer arrays has emerged as a promising modality for short-range hand and finger pose estimation [7][8][9]. Shen et al. [7] demonstrated full-pose estimation using inertial and magnetic sensor fusion in a structured magnetic field, while Kortier et al. [9] showed hand pose estimation by fusing inertial and magnetic sensing aided by a permanent magnet. Parizi et al. [8] presented AuraRing, an electromagnetic finger-tracking system using a ring-worn coil and wrist-mounted sensors.

However, these prior systems typically employ tightly coupled sensor-fusion architectures (e.g., extended Kalman filters over a unified state space) that can be difficult to tune and validate on custom hardware. Moreover, the integration of such tracking systems with robot simulation environments for teleoperation demonstration remains underexplored.

This work addresses these gaps by presenting a **multi-modal hybrid tracking system** in which each sensing modality owns a distinct subset of the tracking state:

- **Magnetic localization** owns local board-frame magnet motion via dipole inversion on a 3x3 magnetometer array.
- **IMU** owns orientation change relative to a startup reference.
- **Fiducial vision** (AprilTag in the current deployment; WhyCon in the planned upgrade) owns global workspace translation.

A combiner node composes these three channels into a robot end-effector command. The system is implemented on custom BLE-enabled wearable hardware and demonstrated through real-time teleoperation of a simulated Franka Panda manipulator in NVIDIA Isaac Sim via ROS 2.

The contributions of this paper are:

1. A **multi-modal sensor-fusion architecture** with explicit ownership decomposition that avoids the complexity of full-state probabilistic estimation while maintaining smooth, robust teleoperation.
2. A **constrained magnetic dipole inverse solver** with two-stage motion gating, moment-scale fitting, and local-minimum recovery for robust board-local position estimation.
3. A complete **hardware–software pipeline** from a custom BLE wristband through ROS 2 to Isaac Sim, demonstrated at a 120 Hz command rate on a deliberate bench-prototype hardware configuration.
4. **Validation evidence** confirming the correct separation of translational and rotational contributions and robustness to transient sensor dropouts, including a direction-agnostic guided hardware evaluation session.
5. A forward-looking **fiducial upgrade path** in which a wristband-mounted WhyCon marker observed by an overhead camera replaces the wrist-mounted AprilTag cube, reducing wearable size without changing the fusion law.

2. Related Work

2.1 Magnetic Tracking for Hand Pose Estimation

Magnetic tracking exploits the predictable spatial decay of magnetic fields to estimate the position and orientation of a permanent magnet relative to a sensor array. The magnetic dipole model provides an analytical forward model relating magnet state to observed field vectors [9][12]. Kortier et al. [9] fused inertial and magnetic measurements using an extended Kalman filter to estimate 3D hand pose, demonstrating sub-centimeter accuracy at short range. Shen et al. [7] extended this to full six-degree-of-freedom (6-DoF) pose estimation with structured magnetic fields. Chen et al. demonstrated MagX, a wearable untethered hand-tracking system using passive magnets and a wrist-worn magnetometer array [27]. These works validate the

feasibility of magnetic tracking for wearable applications but typically require careful calibration and are sensitive to background field disturbances.

2.2 IMU-Based Orientation Estimation

Inertial measurement units combining accelerometers and gyroscopes are widely used for orientation estimation in wearable devices [10]. The LSM6DSO 6-axis IMU provides firmware-level sensor fusion for real-time roll, pitch, and yaw estimation [11]. While IMUs provide reliable orientation tracking, they cannot provide absolute position due to integration drift in accelerometer-based dead reckoning. This fundamental limitation motivates the combination of IMU orientation with complementary position sources.

2.3 Fiducial Vision for Global Localization

AprilTag fiducial markers provide robust pose estimation through detection and Perspective-n-Point (PnP) solving. Multi-tag configurations improve accuracy and reduce ambiguity. AprilTag-based localization has been used extensively in robotics for calibration, navigation, and augmented reality. The tag36h11 family provides a good balance of detection robustness and dictionary size. However, vision-based pose estimation is subject to occlusion, lighting variation, and frame-rate limitations that make it unsuitable as a sole real-time tracking source for teleoperation.

For wearable applications the AprilTag cube used in this work is deliberately bulky: the multi-tag cube ensures at least one face remains visible to the camera in the operator's workspace, but it is ergonomically unsuitable for a finished product. An alternative family of circular fiducials, WhyCon and its encoded variants WhyCode, uses two concentric circles of known parameters and achieves millimeter-level precision with off-the-shelf cameras while remaining computationally very cheap — CPU utilization on the order of single-digit percent at 840x480 and sub-10 ms detection times have been reported [30]. Because a single planar circle is much smaller than a multi-face cube, WhyCon is the natural candidate for a minimized wristband fiducial observed by a fixed overhead camera.

2.4 Hybrid Sensor Fusion

Hybrid inertial-vision tracking has been explored extensively in augmented reality and robotics. Azuma and Bishop demonstrated that combining inertial and vision sensing improves registration accuracy and robustness compared to either modality alone. However, most hybrid fusion systems employ monolithic estimators (e.g., Kalman filters) that couple all state variables. The approach presented here differs by using a **multi-modal architecture** in which each modality owns distinct state variables, avoiding the need for cross-modal covariance estimation and simplifying validation.

2.5 Teleoperation with Simulation and Imitation Learning

Recent work has increasingly leveraged simulation environments for teleoperation research and demonstration. NVIDIA Isaac Sim provides physics-accurate robot simulation with ROS 2 integration, enabling real-to-sim teleoperation workflows [19]. Isaac Lab extends this with imitation-learning capabilities; the Isaac Lab Mimic pipeline uses teleoperation to collect a small number of demonstrations, then automatically generates additional demonstrations by stitching annotated subtask segments, training behavior-cloning policies that can be deployed back to the real robot [31]. The Franka Panda manipulator is widely used as a research platform due to its open control interfaces and simulation support [44]. Recent community work has demonstrated end-to-end Real2Sim pipelines in which a physical master device teleoperates an Isaac Sim robot and the resulting trajectories are then consumed by Isaac Lab Mimic for imitation learning [32]. This positions any accessible wearable teleoperation system, including the one presented here, as a potential front-end for large-scale demonstration collection.

3. Hardware Platform

3.1 System Overview

The hardware platform consists of four major components: (1) a custom BLE wristband containing the microcontroller, IMU, and magnetometer array sensor board; (2) a permanent magnet as the tracked object; (3) a USB camera observing AprilTag markers; and (4) a host computer running the ROS 2 pipeline and Isaac Sim. The wristband communicates wirelessly with the host via Bluetooth Low Energy (BLE), while the camera connects via USB.

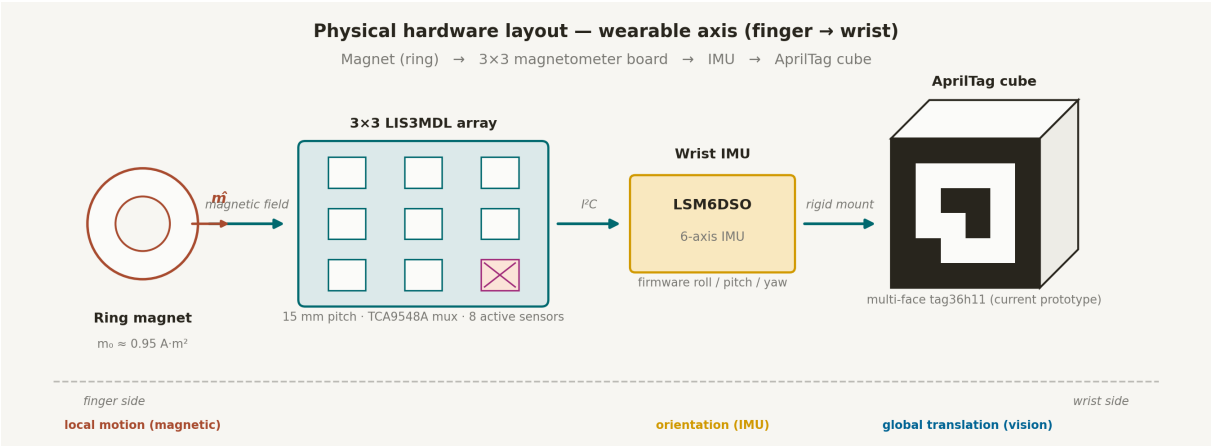

Figure 1. Physical hardware layout along the wearable axis: magnet → 3×3 LIS3MDL array (8 active) → LSM6DSO IMU → AprilTag cube. Each stage owns a different part of the tracking state.

3.2 Deployed Prototype Configuration

A consolidated two-layer PCB was designed in EasyEDA to integrate the STM32WB55 MCU, LSM6DSO IMU, power regulation, and BLE antenna matching network, together with the I2C connectors feeding the 3x3 magnetometer extension board (which additionally carries the TCA9548A multiplexer). During bring-up of the consolidated board a catastrophic connection fault was identified on the TCA9548A side of the extension interconnect. The IMU, MCU, BLE stack, and the remaining half of the extension board continued to function correctly, but the multiplexed 3x3 magnetometer path could not be exercised through the consolidated board.

Rather than halt progress, the demonstrated system runs on a **deliberate bench-prototype configuration** that preserves the full software pipeline while bypassing the faulty interconnect:

- The working half of the consolidated board provides the MCU, BLE link, IMU, and I2C bus fan-out.
- The independently functional 3x3 magnetometer extension board is wired externally to an independently working TCA9548A, and then wired to the MCU through discrete jumper wires.

- All 9 LIS3MDL sensors are reachable through the multiplexer; in the active deployment sensor index 8 is excluded due to a hardware fault on that specific part, leaving 8 active sensors.

The prototype is therefore physically messy — it is a wired bench assembly rather than a finished wearable — but it is functionally complete: every sensing modality described in this paper is exercised in the demonstrated system. The planned consolidated rework is listed in Section 10 as future hardware work.

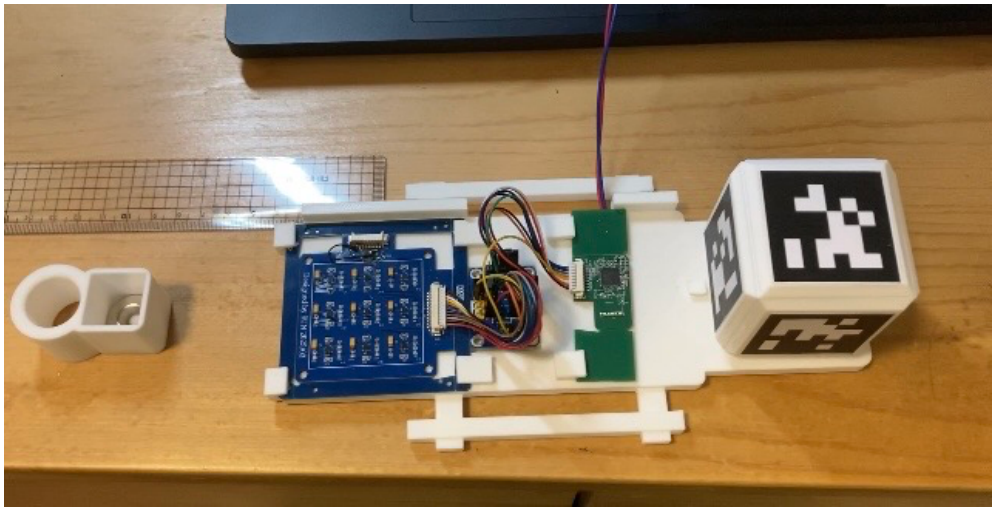

Figure 2. The full hardware prototype.

3.3 Microcontroller Unit

The MCU is an STM32WB55CU6, selected for its dual-core ARM Cortex-M4/M0+ architecture with integrated BLE 5.0 radio, making it suitable for low-power wearable applications. The MCU manages two I2C buses:

- **Software I2C** on PA9 (SCL) and PA10 (SDA) for communication with the IMU.
- **Hardware I2C** on PB8 (SCL) and PB9 (SDA) for communication with the magnetometer array via the I2C multiplexer.

The firmware is developed in STM32CubeIDE with pin configuration generated by STM32CubeMX. Programming and debugging use the Serial Wire Debug (SWD) protocol through exposed test points for SWDIO, SWCLK, VCC, and GND.

The BLE service is configured with:

- Device name: MRC_IMU_BLE
- Service UUID: 0000ff00-cc7a-482a-984a-7f2ed5b3e58f
- Control characteristic UUID: 0000ff01-8e22-4541-9d4c-21edae82ed19
- Data characteristic UUID: 0000ff02-8e22-4541-9d4c-21edae82ed19
- Output data rate (ODR) command code: 6

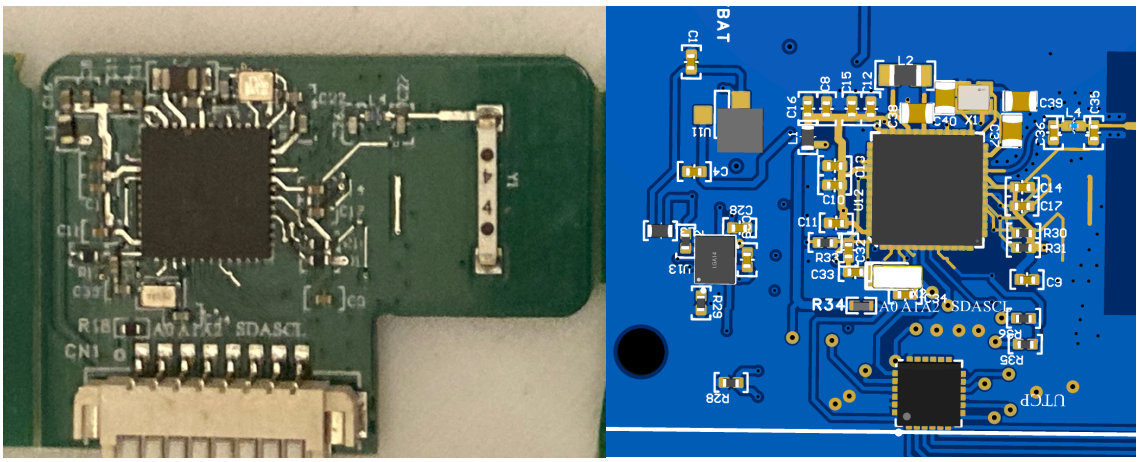

Figure 3. (left): MCU next to the antenna on the physical hardware, (right): MCU 3D model on EasyEDA.

3.4 Inertial Measurement Unit

The IMU is an LSM6DSO from STMicroelectronics, a 6-axis device integrating a 3-axis accelerometer and 3-axis gyroscope [11]. The firmware computes roll, pitch, and yaw and embeds these values in the magnetometer-bearing BLE frames as Q16 fixed-point degrees:

$$\theta_{\text{deg}} = \frac{\text{raw}_{q16}}{65536}$$

Raw sensor conversions are:

$$\omega_{rad/s} = \frac{\text{raw}_{gyro}}{16.4} \cdot \frac{\pi}{180}$$

$$a_{m/s^2} = \frac{\text{raw}_{accel}}{4096} \cdot 9.81$$

The IMU communicates with the MCU over software I2C at the dedicated bus.

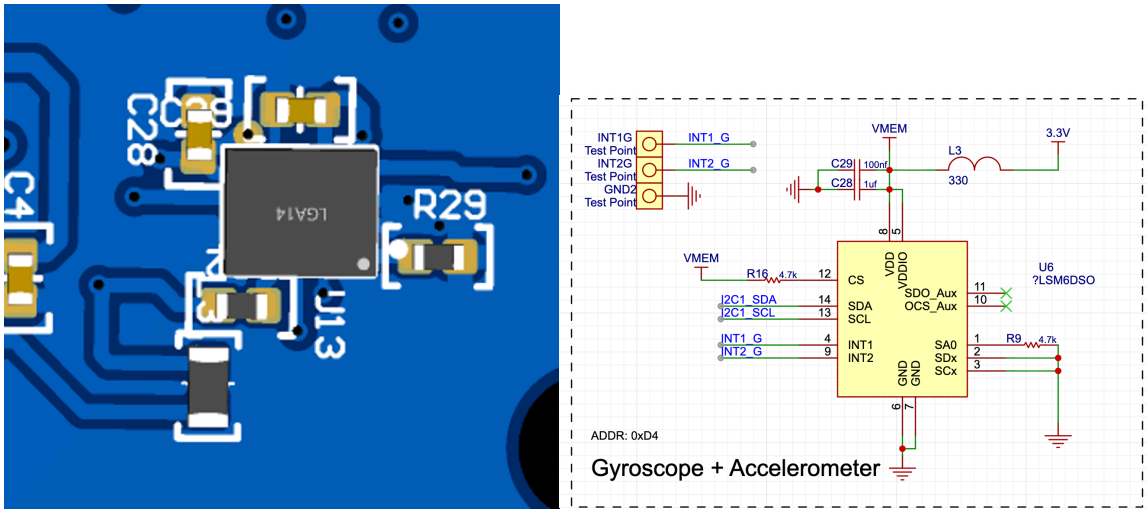

Figure 4. (left): 3D model of the IMU in easyEDA, (right): electric wiring schematic of the IMU on EasyEDA.

3.5 Magnetometer Array

The magnetic sensing subsystem comprises a 3x3 planar array of nine LIS3MDL ultra-low-power 3-axis magnetometers [12]. The sensors are mounted on an extension board with a center-to-center pitch of 15 mm. The physical layout and firmware indexing are:

Row	Col 0	Col 1	Col 2
Top	V5 (idx 0)	G3 (idx 3)	V2 (idx 6)
Middle	G4 (idx 1)	V3 (idx 4)	G1 (idx 7)
Bottom	V4 (idx 2)	G2 (idx 5)	V1 (idx 8)

where **V** denotes a sensor with the SA1 address pin connected to VCC (I2C address 0x1E) and **G** denotes SA1 connected to GND (address 0x1C). This dual-address scheme allows two LIS3MDL sensors to coexist on the same I2C bus segment.

The nine sensors are multiplexed through a TCA9548A 8-channel I2C switch connected to the hardware I2C bus. Each downstream channel of the TCA9548A hosts up to two sensors with distinct addresses, requiring only 5 of the 8 available channels for the 3x3 array. The raw magnetometer conversion is:

$$B_{\mu T} = \frac{\mathrm{raw}_{mag}}{6842} \times 100$$

In the current deployment, sensor index 8 is excluded due to a hardware fault on the individual part, yielding 8 active sensors.

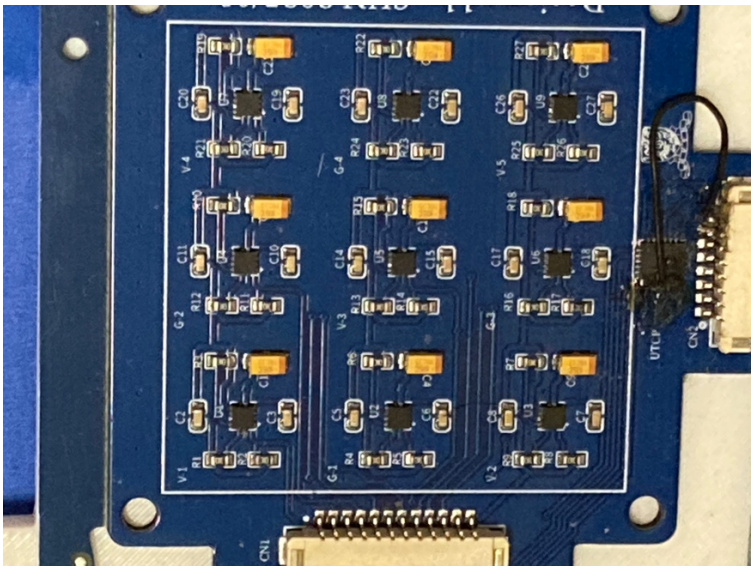

Figure 5. Magnetometer array sensor board on the physical hardware.

3.6 BLE Packet Protocol

The firmware transmits two packet types over BLE:

- **Raw IMU packet** (marker 0xFA, 17 bytes): contains a timestamp and six signed 16-bit little-endian values for gyroscope and accelerometer axes.
- **Magnetometer frame** (marker 0xFE single, 0xFD chunked, 75-byte core): flags, diagnostic bytes, HAL status, timestamp, firmware roll/pitch/yaw in Q16, a sensor-validity mask, and nine 3-axis magnetic field vectors.

The BLE receiver node on the host reconstructs these packets and publishes a unified SensorData ROS message containing:

- float64[27] mag_fields_ut — flattened 9x3 magnetic field in microtesla
- float64[3] gyro_rads — gyroscope in rad/s
- float64[3] accel_mps2 — accelerometer in m/s^2
- float64[3] firmware_rpy_deg — firmware roll, pitch, yaw in degrees
- bool has_firmware_rpy — validity flag

3.7 Camera

The vision subsystem uses a Logitech Brio-class USB camera operating at 1920x1080 resolution and 60 Hz frame rate, accessed through the V4L2 interface. In the current deployment the camera is hand-placed to observe the workspace containing an AprilTag cube mounted on the wristband. The planned overhead-camera variant (Section 10) keeps the same V4L2 interface.

3.8 Tracked Object

The tracked object is a permanent magnet. The magnet is modeled as a magnetic dipole with nominal moment magnitude $m_0 = 0.95\ A \cdot m^2$, oriented approximately along the $-Z$ direction in the board frame (pointing toward the sensor board during typical wearing). The magnet is supposed to imitate a pen's location when a human uses it to "write" while the wristband is worn on the wrist, providing a natural interaction geometry in which the magnet-to-board distance is typically 0.10–0.20 m.

4. Magnetic Localization

4.1 Role in the Hybrid System

The magnetic localization channel estimates the magnet's position **relative to the sensor board**. It does not own global position and does not own end-effector orientation. Its output is interpreted as local motion, which is subsequently rotated by the IMU orientation and added to the global fiducial translation by the hybrid combiner.

4.2 Sensor Geometry

Let the sensor centers be $\mathbf{s}_i \in \mathbb{R}^3$ for $i = 1, \ldots, N$, where $N = 8$ active sensors in the current deployment. The board-plane geometry is generated from the physical grid layout with 15 mm pitch. The board frame is defined with $+Z$ as the board normal.

4.3 Magnetic Dipole Forward Model

For a magnetic dipole at position $\mathbf{p}$ with moment vector $\mathbf{m}$, the field at sensor i is:

$$\mathbf{r}_i = \mathbf{s}_i - \mathbf{p}, \quad \hat{\mathbf{r}}_i = \frac{\mathbf{r}_i}{\| \mathbf{r}_i \|}$$

$$\mathbf{B}_i(\mathbf{p}, \mathbf{m}) = \frac{\mu_0}{4\pi} \frac{3(\mathbf{m} \cdot \hat{\mathbf{r}}_i)\hat{\mathbf{r}}_i - \mathbf{m}}{\| \mathbf{r}_i \|^3}$$

with $\mu_0/4\pi = 10^{-7}$ T·m/A. The implementation computes in SI units (tesla) and converts to microtesla for comparison with measurements.

4.4 State Parameterization

The solver uses a six-dimensional state vector:

$$\mathbf{x} = [x, y, z, m_x, m_y, m_z]$$

subject to the unit-norm constraint $m_x^2 + m_y^2 + m_z^2 = 1$. The effective dipole moment is $m_0\hat{\mathbf{m}}$ where $\hat{\mathbf{m}} = [m_x, m_y, m_z]^T$ and $m_0 = 0.95$ A·m^2. This parameterization separates position estimation from moment-direction estimation, with the moment magnitude handled through a per-frame scaling factor.

4.5 Measurement Preprocessing

The active magnetic pipeline processes each frame through the following stages:

1. **Receive** raw 9x3 magnetic field from SensorData.
2. **Calibrate** using per-sensor calibration vectors if available.
3. **Remap** raw axes using the lis3mdl_user_board_v1 mapping, i.e. model x = raw z, model y = −raw x, model z = −raw y.

4. **Mask** clipped components exceeding the 470 μT threshold.
5. **Drop** failed sensors (index 8).
6. **Subtract** per-sensor background field: $\mathbf{B}_i^{meas} = \mathbf{B}_i^{cal} - \mathbf{B}_i^{bg}$.
7. **Reject** weak-signal frames with peak magnitude below 1.0 μT.

The system intentionally uses per-sensor background subtraction rather than spatial-mean subtraction. This design choice preserves the absolute spatial structure of the dipole field and avoids reintroducing translation–orientation coupling that can occur when the sensor-wise mean is removed at every frame.

4.6 Objective Function

Let W be the residual mask produced by clipped-component handling and λ the per-frame moment-scale factor. The solver minimizes:

$$J(\mathbf{p}, \hat{\mathbf{m}}, \lambda) = \left\| W \odot \left(\mathbf{B}^{meas} - \lambda\, \mathbf{B}^{pred}(\mathbf{p}, m_0 \hat{\mathbf{m}}) \right) \right\|_F + w_{pos} \parallel \mathbf{p} - \mathbf{p}_{prior} \parallel^2$$

subject to $\parallel \hat{\mathbf{m}} \parallel = 1$ and box constraints on position. The moment-scale factor λ compensates for variations in effective moment magnitude due to temperature, manufacturing tolerance, or distance-dependent model error. It is computed by projecting the nominal predicted field onto the measured field and clipping to $[\lambda_{\min}, \lambda_{\max}] = [0.75, 1.20]$. The position regularization weight is $w_{pos} = 0.3$.

4.7 Solver Strategy

The optimization is solved using Sequential Least Squares Programming (SLSQP) with:

- **State variables**: $[x, y, z, m_x, m_y, m_z]$
- **Equality constraint**: unit-norm on moment direction
- **Box constraints**: $x \in [-0.20, 0.20]$ m, $y \in [0.02, 0.30]$ m, $z \in [-0.20, 0.20]$ m
- **Warm start**: previous frame's solution
- **Initial guess** (first frame): $[0.0, 0.10, 0.0, 0.0, 0.0, 1.0]$
- **Iteration budget** in the current hybrid hardware launcher: primary=200, orientation=200, warm_probe=200

4.8 Two-Stage Motion Gating

A key innovation is the two-stage acceptance rule that prevents false translation during pure tilt of the magnet. When a previous state is available:

Stage 1 (orientation-only solve): The previous position is frozen and only the dipole direction is optimized:

$$\hat{\mathbf{m}}_{ori} = \arg\min_{\|\hat{\mathbf{m}}\|=1} J\left(\mathbf{p}_{prior}, \hat{\mathbf{m}}, \lambda\right)$$

Stage 2 (full solve): Both position and direction are free. The full solve is accepted only if its field-only cost improves beyond a threshold:

$$J_{full,field} < (1-\rho)\, J_{ori}$$

with $\rho = 0.10$. If the full solve does not sufficiently improve over the orientation-only explanation, the system concludes that the field change is better explained by tilt than by translation, and the position is held.

4.9 Local-Minimum Recovery

The magnetic dipole inverse problem is known to exhibit local minima, particularly left–right mirror ambiguities for planar sensor arrays. The solver addresses this with mirrored probe seeds around the warm-start solution. If the ordinary warm-start result is not sufficiently good, the solver evaluates laterally mirrored seeds and accepts an alternative only if it beats the current basin by a required margin. This is essential for preventing the solver from becoming trapped in a centered basin during lateral motion.

4.10 Guard Rails and Output

Additional robustness measures include a maximum accepted position jump of 0.020 m per frame, cost-spike rejection factor of 4.0 relative to the running cost, clipped-component masking at 470 μT (the LIS3MDL saturation region), and state smoothing alpha of 1.0 (direct pass-through in the current deployment). The magnetic localizer publishes magnet_pose as a PoseStamped message. The position encodes the estimated magnet's location in the board frame.

The orientation is derived from the recovered dipole direction for visualization and debugging, but only the position is used by the downstream hybrid combiner.

5. IMU Orientation Channel

5.1 Role in the Hybrid System

The IMU is the **sole orientation authority** in the deployed hybrid controller. It does not contribute to translational estimation, and the system does not employ inertial dead reckoning for position.

5.2 Firmware Orientation Construction

The firmware provides roll (ϕ), pitch (θ), and yaw (ψ) in degrees. The runtime converts to a rotation matrix using intrinsic XYZ Euler composition:

$$R_{imu}(t) = R_z(\psi)\, R_y(\theta)\, R_x(\phi)$$

where angles are first converted from degrees to radians.

5.3 Startup-Referenced Delta Orientation

Let $R_{imu}(t_0)$ be the first valid IMU orientation captured by the combiner. The deployed controller uses the delta rotation:

$$R_\Delta(t) = R_{imu}(t)\, R_{imu}(t_0)^T$$

The IMU channel therefore contributes **change relative to startup**, not an absolute global orientation. The corresponding quaternion is:

$$q_\Delta(t) = \mathrm{quat}\big(R_\Delta(t)\big)$$

using the Hamilton scalar-first convention $q = [w, x, y, z]$.

5.4 Design Rationale

The startup-referenced orientation path was retained over a fully coupled inertial-navigation approach because (a) it avoids accelerometer drift accumulation in translation, (b) it is simpler to

validate on custom hardware, and (c) the firmware-level sensor fusion in the LSM6DSO already provides reasonable orientation estimates without requiring an additional estimator. An MEKF implementation exists in the repository but is not exercised in the active hardware launch.

6. Fiducial Vision Channel

6.1 Role in the Hybrid System

The fiducial vision path owns **global translation only** in the final controller. Although the upstream detector estimates a full 6-DoF pose, the working pipeline explicitly strips continuous fiducial orientation before fusion.

6.2 Detection and PnP Pipeline

The AprilTag stack operates as follows:

1. Capture frames from the V4L2 camera at 1920x1080, 60 Hz.
2. Detect tag36h11 markers using the OpenCV-based detector backend with quad-decimation factor 2.0.
3. Solve a multi-tag PnP problem for the cube configuration.
4. Generate two pose candidates per frame: a fresh solvePnPRansac estimate, and a continuity-biased solvePnP estimate seeded from the previous pose.
5. Select the better candidate using a continuity-aware composite score.
6. Apply temporal smoothing and publish.

6.3 Continuity Score

When a previous pose exists, each candidate is scored by:

$$\text{score} = e_{reproj} + 40.0\,\Delta p + 0.03\,\Delta\theta_{\text{deg}}$$

where e_{reproj} is the reprojection RMS error in pixels, Δp is the position jump in metres, and $\Delta\theta_{\text{deg}}$ is the quaternion angular difference in degrees. This scoring function directly encodes a preference for geometrically plausible, temporally smooth poses.

6.4 Guard Rails and Temporal Filtering

Pose rejection thresholds: maximum reprojection RMS of 8 px, maximum position jump of 0.15 m. Temporal filtering: position EMA alpha of 0.7, outlier rejection at 0.10 m, hold-last-pose timeout of 1.0 s.

6.5 Translation-Only Relay

Before the hybrid combiner receives the fiducial stream, an apriltag_translation_only_relay node intercepts each message and replaces the orientation with the identity quaternion:

$$\mathbf{p}_{out} = \mathbf{p}_{tag}, \quad q_{out} = [1,0,0,0]$$

This ensures the hybrid combiner receives fiducial data as a pure translation source, preventing orientation conflicts with the IMU channel.

6.6 Representative Translation Frame

The AprilTag node exports translation in a representative frame with right = -camera_x, up = -camera_y, forward = -camera_z, implemented via axis signs $[-1, -1, -1]$.

7. Hybrid Fusion Law

7.1 Design Philosophy

The central contribution of this work is the **ownership-based multi-modal fusion**. Rather than estimating a single joint state from all sensor measurements through a monolithic filter, the system assigns each modality a specific role and composes contributions algebraically. This avoids the need for cross-modal covariance estimation and makes the system behavior transparent and independently testable.

7.2 Variables

Let:

- $\mathbf{p}_{home} = [0.4, 0.0, 0.35]$ — home end-effector position in the Franka base frame
- $q_{home} = [0.0, 1.0, 0.0, 0.0]$ — home end-effector orientation (180° about x)

- $R_\Delta(t)$ — startup-referenced IMU delta rotation
- $\Delta\mathbf{p}_{mag}^{board}(t)$ — magnetic board-frame magnet-position delta
- $\Delta\mathbf{p}_{tag}^{repr}(t)$ — fiducial representative-frame translation delta

7.3 Axis Remapping

The magnetic and fiducial channels deliver data in their native frames. The combiner applies axis remaps to convert to robot-base semantics:

- **Magnetic remap**: axis map yxz, signs $[1, -1, 1]$, scale $[3, 3, 3]$
- **Fiducial remap**: axis map zxy, signs $[1, -1, 1]$, scale $[1, 1, 1]$

The magnetic scale factor of 3x amplifies the relatively small board-local magnet displacements into meaningful workspace motion.

7.4 Position Fusion Law

Define the remapped fiducial delta $\Delta\mathbf{p}_{tag}^{robot}(t) = \mathcal{M}_{tag}\left(\mathbf{p}_{tag}(t) - \mathbf{p}_{tag}(t_0)\right)$ and the remapped magnetic delta $\Delta\mathbf{p}_{mag}^{robot}(t) = \mathcal{M}_{mag}\left(\mathbf{p}_{ring}(t) - \mathbf{p}_{ring}(t_0)\right)$. The magnetic local motion is rotated into the world frame by the IMU delta:

$$\Delta\mathbf{p}_{mag}^{world}(t) = R_\Delta(t)\,\Delta\mathbf{p}_{mag}^{robot}(t)$$

The final position command is:

$$\mathbf{p}_{cmd}(t) = \mathbf{p}_{home} + \Delta\mathbf{p}_{tag}^{robot}(t) + \Delta\mathbf{p}_{mag}^{world}(t)$$

This is the defining architectural property: **fiducial translation is added directly in the robot frame, while magnetic local motion is first rotated by the IMU orientation change.**

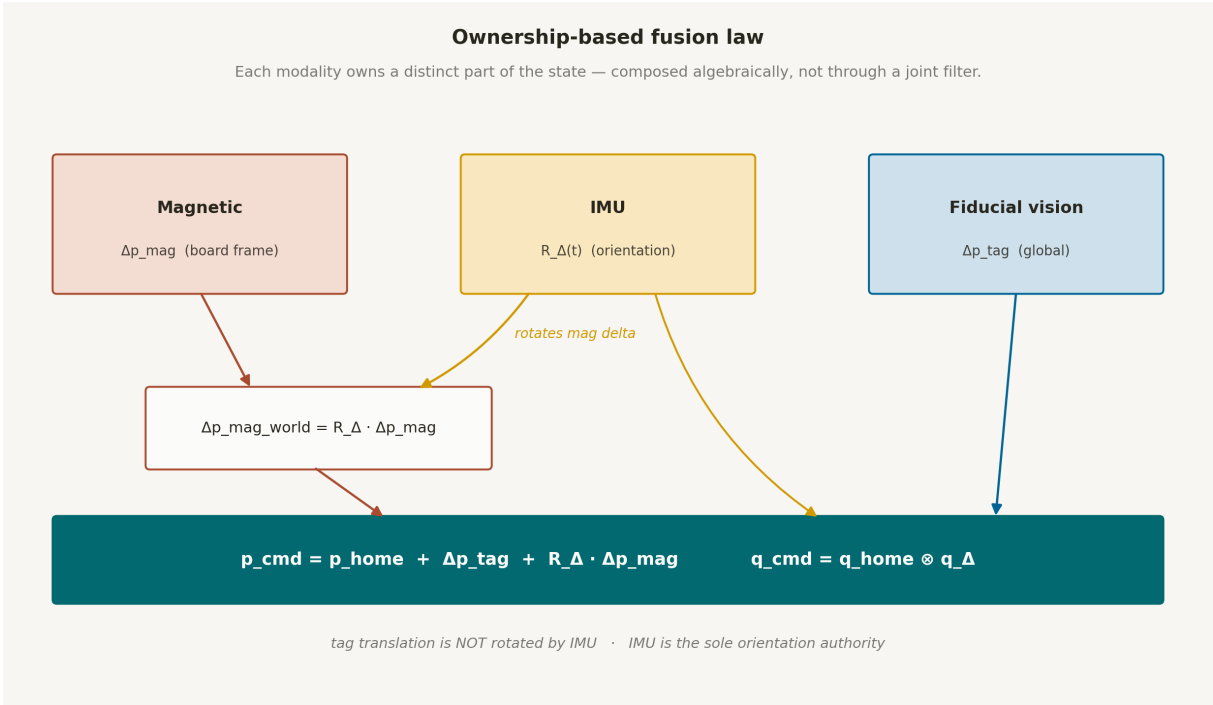

Figure 6. Ownership-based fusion law. Magnetic owns local board-frame motion, IMU owns orientation, vision owns global translation; the IMU rotates the magnetic delta but never the tag translation.

7.5 Orientation Fusion Law

The commanded orientation is $q_{cmd}(t) = q_{home} \otimes q_{\Delta}(t)$, where $\otimes$ denotes Hamilton quaternion multiplication. The end-effector therefore follows the operator's wrist rotation relative to the startup pose.

7.6 Availability and Dropout Model

Each channel is treated independently. Before a channel registers its first reference, its contribution is zero. After registration, stale data does not invalidate the command: the combiner holds the last contribution from each channel and continues publishing valid = true. This dropout model is critical for smooth operation — the system publishes a valid home pose from the very first timer tick, avoiding the failure mode where the robot waits indefinitely for all sensors.

7.7 Smoothing and Workspace Clipping

The combiner supports EMA smoothing for position and orientation. In the current deployment, both smoothing alphas are set to 1.0 (direct pass-through). For orientation smoothing when enabled, quaternion hemisphere alignment prevents sign-ambiguity jumps. The commanded position is clipped to the workspace bounds $[0.28, -0.30, 0.12]$ to $[0.60, 0.30, 0.55]$ m before publication.

8. System Integration with Isaac Sim

8.1 ROS 2 Architecture

The complete system runs on ROS 2 Jazzy with the rmw_fastrtps_cpp middleware. The active node graph is:

1. ble_receiver — BLE interface publishing sensor_data
2. matlab_localizer — magnetic inverse solver publishing magnet_pose
3. apriltag_fusion — camera detection publishing cube_pose_raw
4. apriltag_translation_only_relay — orientation stripper publishing cube_pose
5. magnet_imu_tag_combiner — hybrid fusion publishing franka_target and board_orientation
6. isaacsim_bridge — simulator interface publishing target_pose

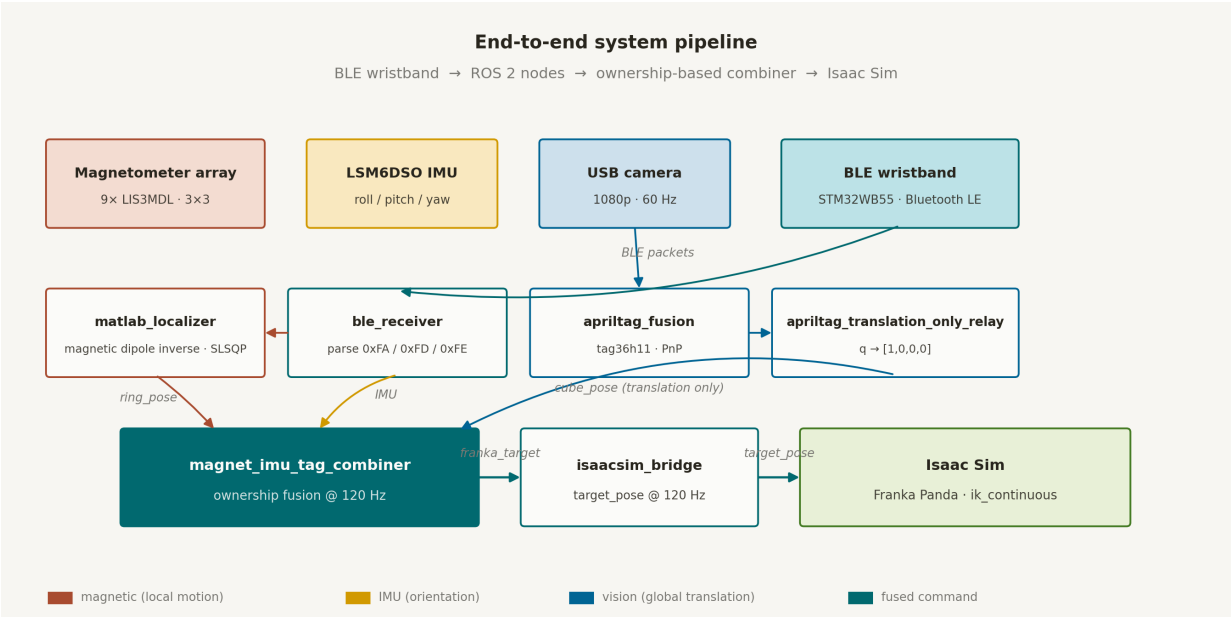

Figure 7. End-to-end pipeline. Sensors feed ROS 2 nodes that are composed by the ownership-based combiner at 120 Hz into the Isaac Sim Franka Panda controller.

8.2 Isaac Sim Scene

The simulator runs NVIDIA Isaac Sim with a Franka Panda manipulator. The bridge publishes target_pose at 120 Hz. The default control mode is ik_continuous targeting the panda_hand frame, providing smooth inverse-kinematics-based tracking of the commanded end-effector pose. A twist feedforward path exists in the codebase but is disabled in the active hardware deployment.

8.3 Startup Sequence

A readiness gate in the launch script waits for fresh signals from all channels (fiducial translation, IMU orientation, magnet pose, fused target, and target pose) before launching the Isaac Sim scene. This ensures the simulator only becomes active after the full sensing chain is live.

8.4 Timing

Component	Rate
BLE sensor publication	~60 Hz

Component	Rate
AprilTag detection	60 Hz
Hybrid combiner	120 Hz
Isaac bridge	120 Hz

9. Validation and Results

The system tracked end-effector position and orientation in real time throughout all tests, with the simulated Franka arm following operator motion smoothly and without freezing. Validation was conducted at two levels: a unit-level software test of the fusion combiner and a guided live hardware evaluation session. Across both levels the system demonstrated correct ownership decomposition — fiducial translation moved the end-effector without affecting orientation, IMU tilt changed orientation without translational leakage, and local magnetic motion was correctly rotated by the IMU into the world frame. Startup stability was confirmed with cube drift below 5.7 mm and a fused output rate of 239.9 Hz on the fused command, and the system robustly handled transient fiducial occlusions of up to 1.167 s without losing control continuity.

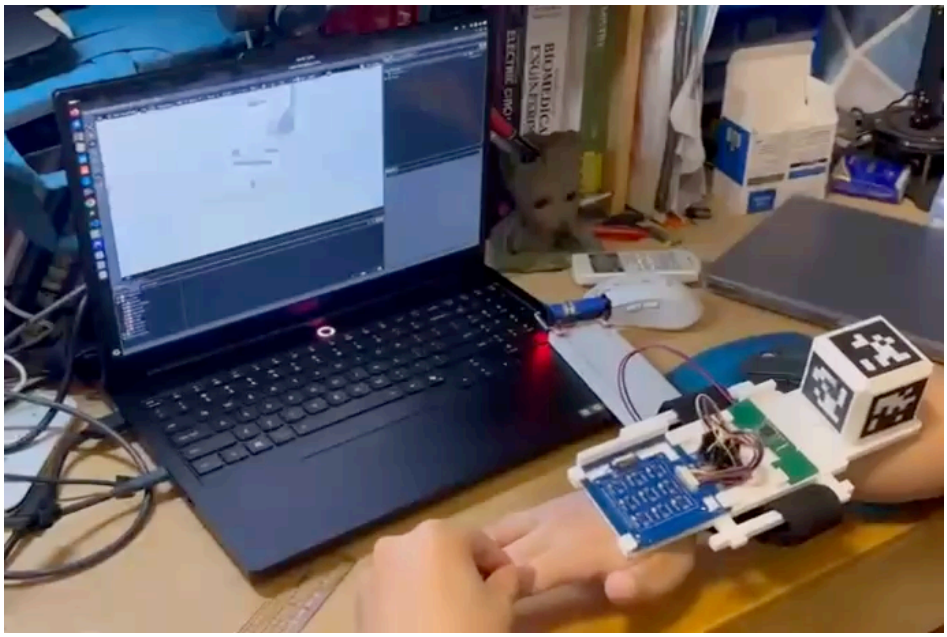

Figure 8. Full testing workspace with IsaacSim for real-time teleoperation.

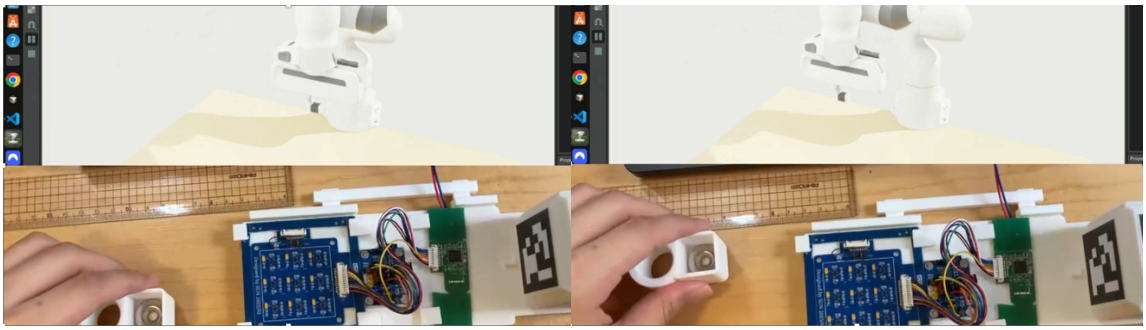

Figure 9. Real-time magnet tracking results.

9.1 Unit-Level Combiner Validation

A focused test suite (test_magnet_imu_tag_combiner.py) directly validates the deployed fusion law by injecting synthetic sensor data and checking the combiner output. The validated behaviors are:

1. Fiducial-only translation changes end-effector position without affecting orientation.
2. IMU-only tilt changes orientation without affecting the fiducial global offset.
3. IMU tilt does not rotate the fiducial offset — confirming that global translation is independent of board attitude.
4. Magnetic local delta is rotated by the IMU and added on top of the fiducial offset — confirming the position fusion law.

These tests provide strong evidence for the correctness of the ownership decomposition because they verify the combiner arithmetic directly rather than relying solely on end-to-end observation.

9.2 Guided Hardware Evaluation

A guided isolated hardware evaluation session analyzed each channel in a direction-agnostic way, emphasizing movement magnitude, path length, continuity gaps, and orientation excursion rather than assuming a specific axis sign was executed perfectly. The goal was to show whether each sensing channel exhibited the correct qualitative ownership.

Selected measured results:

Claim	Evidence
Startup stability	Cube drift 5.7 mm; /franka_target at 239.9 Hz; /target_pose at 120.0 Hz
Fiducial translation ownership	/cube_pose path length 0.370 m, max displacement 119.6 mm; relay orientation held at 0.00° while raw pose moved 9.10°
Magnetic local motion	/ring_pose path length 0.144 m, max displacement 77.5 mm, z-range only 3.8 mm in flat trial
IMU orientation ownership	/board_orientation reached 41.0° orientation excursion with zero translational leakage
Vision dropout recovery	2 gaps >0.5 s, worst gap 1.167 s, then resumed
Tilted magnetic trial	Invalid (board angle only 1.00°, magnet motion 0.0 mm) — requires recollection

Observed per-channel rates in this session were consistent with the current pipeline split: AprilTag translation ~54 Hz, IMU orientation ~35 Hz, magnetic magnet_pose ~14 Hz, fused /target_pose ~120 Hz.

The session revalidated AprilTag translation-only ownership, nonzero flat magnetic local motion, IMU-only orientation response, and AprilTag dropout recovery. It did **not** revalidate the tilted-magnetic-motion rotation claim because the tilt segment was not executed strongly enough to produce a useful signal. That specific test requires recollection.

9.3 Hardware Runtime Behavior

Additional qualitative behaviors observed in the deployed demonstration:

- **Flat magnetic circles**: When the board is held flat, circular finger motion produces circular end-effector motion in the workspace.
- **Global translation**: A fiducial forward translation shifts the workspace origin without deforming the local magnetic motion pattern.
- **Rotated local motion**: After a 90° IMU tilt, the same local magnetic motion is expressed in a rotated world direction, including motion into the robot Z direction. This is qualitatively observed but is the item flagged for quantitative recollection above.

- **Dropout resilience**: Transient fiducial occlusions do not invalidate the target stream or freeze the robot arm.

9.4 Latency Decomposition

The deployed pipeline can feel visibly latent, but the dominant latency source is the magnetic solver rather than downstream republishing:

$$T_{mag,e2e} \approx T_{BLE\ frame} + T_{preprocess} + T_{solve} + T_{combiner} + T_{bridge} + T_{sim}$$

The active hybrid launcher still runs the MATLAB-faithful SLSQP solver at the conservative 200/200/200 iteration budget (primary, orientation-only gate, warm-probe), while a separate low-latency magnet-only wrapper uses 60/40/30. The downstream combiner and bridge already operate at 120 Hz with no additional EMA lag, and the hot ROS path uses queue depth 1 throughout. AprilTag adds its own deliberate translation smoothing (EMA alpha 0.7, hold-last-pose 1.0 s), but this affects the global translation channel, not the solver-bound lag of the magnetic path.

10. Discussion and Future Work

10.1 Advantages of partitioned Fusion

The ownership-based architecture offers several advantages over monolithic sensor-fusion approaches: **transparency** (each channel's contribution is independently observable and testable), **modularity** (channels can be added, removed, or replaced without redesigning the estimator), **robustness** (the independent dropout model means graceful degradation), and **simplicity** (the absence of cross-modal covariance estimation eliminates a major source of tuning difficulty).

10.2 Limitations

Several limitations should be acknowledged:

1. **Bench-prototype hardware**: The demonstrated system runs on a deliberate wired assembly that bypasses the faulty consolidated PCB interconnect. Although every

modality described in this paper is exercised correctly, the current physical form factor is not a finished wearable.

2. **No full-state probabilistic estimator**: The system does not optimally weight measurements based on uncertainty. A tightly coupled filter could in principle achieve better accuracy by exploiting statistical correlations.
3. **No external ground-truth evaluation**: The validation in this work focuses on ownership correctness, stability, and dropout resilience, without an external motion-capture or instrumented-robot reference due to an expensive cost; a full quantitative pose-error evaluation is left as future work.
4. **Euler parameterization**: The firmware RPY path using intrinsic XYZ Euler angles is subject to gimbal lock near singular configurations, though the quaternion output from the combiner itself avoids this issue.
5. **Rigid offset assumption**: The current deployment assumes zero rigid offset between the AprilTag cube and the reference point. Physical misalignment should be measured and compensated.
6. **Background field sensitivity**: The magnetic localizer requires accurate per-sensor background field calibration, performed without the magnet present. Environmental magnetic disturbances can degrade localization accuracy.
7. **Single faulty sensor**: Sensor index 8 is currently excluded, reducing the array to 8 active elements. A replacement part would restore the full 3x3 configuration.
8. **AprilTag cube bulk**: The wrist-mounted AprilTag cube is ergonomically unsuitable for a finished wearable. This is addressed directly in Section 10.3.

10.3 Planned Fiducial Upgrade: Overhead Camera with WhyCon

The most impactful near-term upgrade is the replacement of the wrist-mounted AprilTag cube with an **overhead camera** observing a small **WhyCon** circular fiducial mounted flat on the wristband. The motivation and properties of this change are as follows.

Why it matters for the wearable. The current AprilTag cube provides multi-face visibility at the cost of ergonomic bulk. A single planar circular fiducial is intrinsically much smaller than a cube and can be laminated directly onto the wristband enclosure. Replacing the cube eliminates

the dominant physical protrusion on the current prototype and is the single largest step toward a finished wearable form factor.

Why the fusion law does not change. The role of the vision channel in this paper is strictly **global translation**. WhyCon, like AprilTag, provides millimeter-precision planar detection with off-the-shelf cameras at high frame rate [30]. Because the translation-only relay in Section 6.5 already strips continuous orientation from the fiducial stream before fusion, a circular marker (WhyCon) that natively provides a strong positional estimate fits the existing architecture without any change to the combiner. WhyCode, the encoded variant, can optionally be used if multiple wristbands need to be distinguished in a multi-user experiment.

Why overhead is the natural viewing geometry. A fixed overhead camera removes the need to hand-place a forward-facing camera and produces a clean top-down projection in which planar fiducials are maximally observable. It also provides a stable workspace frame for the global translation channel and is trivially extensible to multiple simultaneous operators. Crucially, the existing apriltag_translation_only_relay and the combiner's fiducial remap (axis map zxy, signs $[1, -1, 1]$) are the only pieces that need re-tuning; the rest of the pipeline is modality-agnostic.

Net effect. The planned configuration substantially reduces wearable size while maintaining the translation-ownership role of the vision channel, and it requires no change to the magnetic or IMU paths.

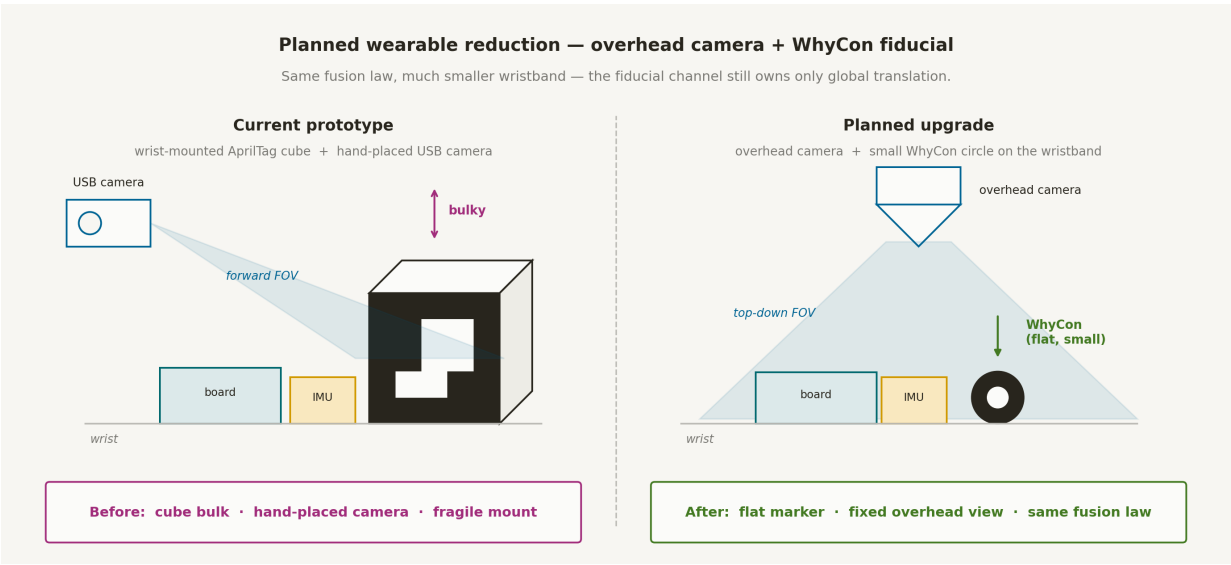

Figure 8. Planned wearable reduction. An overhead camera observes a small WhyCon circle directly on the wristband, replacing the bulky AprilTag cube while leaving the fusion law unchanged.

10.4 Consolidated PCB Rework

Section 3.2 described the deliberate bench-prototype used in this work. The corresponding hardware future-work item is a respin of the consolidated PCB with corrected TCA9548A interconnect routing, replacement of the faulty individual sensor (index 8) so that the full 9-sensor 3x3 array is live, and integration into a 3D-printed wrist enclosure that co-locates the MCU, IMU, and magnetometer array together with the WhyCon fiducial on the top surface.

10.5 Haptic-Feedback Ring and 4x4 Array

A BLE-enabled ring containing a vibration motor would close the teleoperation loop by driving tactile feedback from simulated contact events in Isaac Sim. The existing BLE stack on the wristband already supports the packet budget for a second BLE peripheral. A secondary improvement is expanding the magnetometer array from 3x3 to 4x4 for improved spatial resolution and reduced local-minimum ambiguity in the dipole inverse.

10.6 Real2Sim and Imitation Learning

A second forward-looking direction treats the entire wearable-plus-simulator stack as a **Real2Sim demonstration collection platform**. The Franka Panda in Isaac Sim already consumes end-effector commands at 120 Hz from the hybrid combiner. Isaac Lab and the Isaac Lab Mimic pipeline provide a standard path from recorded teleoperation trajectories to behavior-cloning policies: a small number of human demonstrations are automatically expanded into a larger synthetic dataset by stitching annotated subtask segments, and the resulting dataset is used to train state-based or visuomotor BC policies [31]. Recent community pipelines have demonstrated this end-to-end with ROS 2 as the connector between a physical master device and Isaac Sim, followed by Isaac Lab Mimic for policy training [32]. Integrating the wearable presented here with this pipeline would convert it from a pure teleoperation demo into a **low-cost demonstration-collection front-end for imitation learning**, targeting tasks such as cube stacking, pick-and-place, and simple assembly.

10.7 Transfer to a Real Franka

A logical continuation is closed-loop operation on a physical Franka Panda (not only the simulator) through libfranka or franka_ros2. The existing target_pose topic is already the interface used to drive Isaac Sim and can be consumed directly by a real-robot IK controller. Known sim-to-real gaps must be addressed: the simulator ik_continuous controller runs at a fixed 120 Hz, whereas a real Franka typically accepts Cartesian-impedance or joint-velocity commands at 1 kHz; camera calibration and workspace registration must be redone for the physical workspace; and workspace clipping in the combiner must match the real robot's safe volume. A policy trained in simulation via the Isaac Lab Mimic path of Section 10.6 can be deployed on the real arm, with sim-to-real issues localized to the policy execution layer rather than the tracking front-end.

10.8 Fully Coupled Estimation as an Option

For completeness, a tightly coupled MEKF or factor-graph estimator over all three modalities could be investigated as an alternative to the multi-modal architecture. The existing MEKF implementation in the codebase is the natural starting point. The intention is not to replace the partitioned controller but to provide a directly comparable baseline for future accuracy studies.

11. Conclusion

This paper presented a **mutli-modal hybrid tracking system** for low-cost robot teleoperation that deliberately separates sensing responsibilities: magnetic localization for board-local motion, IMU for orientation, and fiducial vision for global translation. The constrained magnetic dipole inverse solver with two-stage motion gating and local-minimum recovery provides robust board-frame position estimation. The hybrid combiner composes these channels at 120 Hz without requiring cross-modal covariance estimation, and the dropout-hardened architecture maintains smooth operation through transient sensor failures. The system is implemented on a deliberate bench-prototype hardware configuration — a consequence of an unresolved fault on the consolidated PCB — and is demonstrated through real-time teleoperation of a simulated Franka Panda manipulator in NVIDIA Isaac Sim via ROS 2. A guided hardware-evaluation session validated startup stability, fiducial translation ownership, magnetic local motion, IMU

orientation response, and vision-dropout recovery; it also correctly flagged the tilted-magnetic-rotation test as needing recollection. Planned near-term work replaces the bulky AprilTag cube with a small WhyCon fiducial observed by an overhead camera, substantially reducing the wearable's size while preserving the translation-ownership role. The results validate the ownership decomposition as a practical alternative to monolithic sensor fusion for accessible teleoperation systems and position the platform as a natural demonstration-collection front-end for imitation-learning workflows in Isaac Lab Mimic and for eventual deployment on a physical Franka Panda.

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